

KAMIL TYLUS

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SUMMARY

Data Scientist with 2+ years of experience building production ML and data systems. Strong software engineering foundation in Python, testing, and system design. Experience building and deploying LLM applications with agentic workflows, RAG, evaluation, and AWS/Terraform.

PROJECTS

Chess Opening Assistant — [link](#)

May 2026 – Present

Built and deployed an AI assistant for learning chess openings, combining retrieved theory, engine analysis, and LLM reasoning in a stateful workflow.

- Designed RAG over 300 annotated chess positions using exact board-state lookup instead of semantic embeddings, matching retrieval directly to the structure of the domain.
- Built a LangGraph workflow orchestrating retrieval, tool execution, and response generation, using engine analysis of the current position to reduce hallucinations and losing lines.
- Created a labeled evaluation suite combining deterministic tool-use checks with LLM-as-a-judge answer scoring, with LangSmith for tracing and evaluation runs. Added Redis caching for repeated tool calls on the same position.
- Containerized the FastAPI application and deployed it on AWS using Terraform for infrastructure as code, with automated testing through GitHub Actions. Added rate limiting and budget safeguards to guard against abuse and uncontrolled LLM costs.

EXPERIENCE

Data Scientist at Verisk

Jul 2024 – Present

- Developed a basement-presence prediction model over UK property data with sparse positive labels, engineering geospatial features and designing the modeling and validation approach around incomplete ground truth.
- Refactored a ~15k LOC rebuild cost estimation codebase: introduced schema validation, added test coverage to a previously untested system, and consolidated duplicate processing routes — reducing bug surface and improving maintainability.

EDUCATION

Master's degree in Computer Science (ML/AI Specialization), Jagiellonian University

2023 – 2025

Graduated with distinction (GPA: 4.92/5.0), Rector's Scholarship

- **Thesis:** "Explanations through molecular subgraph contributions in molecular property prediction" — GNN explainability for molecular property prediction.

Bachelor's degree in Computer Science, Jagiellonian University

2020 – 2023

Graduated with distinction (GPA: 4.89/5.0), Rector's Scholarship

SKILLS

Languages:

Python, SQL

LLM / AI Eng:

LangGraph, LangChain, LangSmith, RAG, agentic systems, LLM evaluation

ML:

PyTorch, Scikit-learn, XGBoost, CatBoost, Optuna, GNNs

Data:

Polars, Pandas, GeoPandas

Engineering / MLOps:

FastAPI, Docker, AWS, Terraform, Redis, GitHub Actions

Spoken:

Polish (Native), English (C1)